



Design Criteria for Film Cooling for Small Liquid-Propellant Rocket Engines

R. CARL STECHMAN,* JOELEE OBERSTONE,† AND J. C. HOWELL‡

The Marquardt Corporation, Van Nuys, Calif.

Results of a study to determine the applicability of film cooling to rocket engines in the 10- to 1000-lb-thrust range using earth-storable, space-storable, and cryogenic propellant combinations are presented. The accuracy of the analytical model was verified with test data using the propellant combinations of nitrogen tetroxide/monomethylhydrazine and fluorine/monomethylhydrazine (N_2O_4/MMH and F_2/MMH). Data are presented which illustrate the inter-relational effects of mixture ratio, specific impulse, wall temperature, and the percent of film cooling. The studies show that N_2O_4/MMH and ClF_3/MMH and other similar earth-storable propellant combinations are readily adaptable to small film-cooled spacecraft engines. The space-storable and deep-cryogenic systems OF_2/MMH , OF_2/B_2H_6 , OF_2/CH_4 (LPG's), F_2/MMH , O_2/H_2 , and F_2/H_2 will require the use of film cooling and high-temperature materials if the potentially high performance they possess is to be realized. Otherwise, the systems have to be operated fuel rich with high percentages of fuel film cooling, which will compromise the deliverable specific impulse and system weight. For example, if a 100-lb-thrust F_2/H_2 engine operating at 100-psia chamber pressure was needed to provide 400 sec of I_{sp} at a mixture ratio of 8.0 or greater, the minimum percent of film cooling and minimum wall temperature capability of the chamber material must be 40% and 5000°F, respectively.

Nomenclature

A = area
 C_f = friction coefficient
 C_p = heat capacity at constant pressure
 D = diameter of engine
 H = total enthalpy
 h = heat-transfer coefficient
 I_{sp} = specific impulse
 k = thermal conductivity
 L = length of liquid section
 P = chamber perimeter
 Pr = Prandtl number = $\mu C_p / k$
 Q = heat flux
 r = chamber radius
 Re = Reynolds number = $\rho u X / \mu$
 S = initial thickness of gaseous film
 St = Stanton number = $h / \rho u C_p$
 T = total temperature
 u = velocity
 W = flow rate
 X = distance downstream

α = thermal diffusivity
 δ = liquid film thickness
 η = dimensionless liquid coolant efficiency factor
 λ = heat of vaporization
 μ = viscosity
 ρ = density
 σ = dimensionless factor accounting for variation of ρ and μ values across boundary layer
 τ = shear stress

Subscripts

c = condition of gaseous coolant
 f = average film property
 g = condition of main core gas
 eff = gas at wall
 I = initial condition
 L = condition of liquid coolant
 r = recovery condition
 s = condition at vaporization temperature
 w = wall condition
 x = condition based on downstream distance
 $*$ = nozzle throat condition

Presented as Paper 68-617 at the AIAA 4th Propulsion Joint Specialist Conference, Cleveland, Ohio, June 10-14, 1968; submitted June 17, 1968; revision received October 30, 1968.

* Manager, Development Engineering, Rocket Systems Division. Member AIAA.

† Member of Advanced Technical Staff, Advanced Technology, Rocket Systems Division; now Member of Technical Staff, Applied Mechanics Division, Aerospace Corporation. Associate Member AIAA.

‡ Member of Advanced Technical Staff, Technology, Rocket Systems Division. Associate Member AIAA.

Introduction

THERE is a strong interest in the application of advanced propellant combinations to small rocket engines. Most of the advanced propellant technology gathered to date has been with engines in the 1000- to 100,000-lb-thrust level. Until recently, little was known about the characteristics of these propellant combinations when used in small rocket engines, since the technology obtained from the higher thrust level

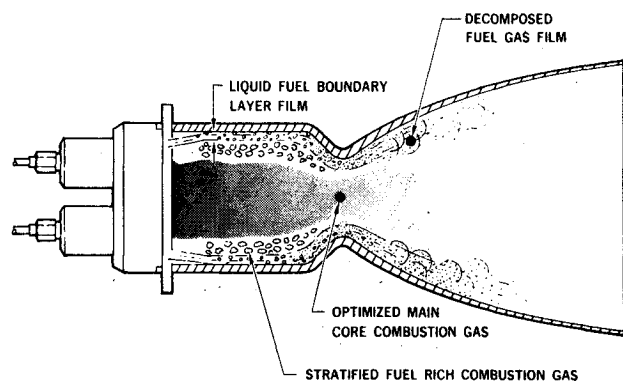


Fig. 1 Film-cooled rocket engine.

engines is not readily adaptable to the smaller engine systems. The high combustion temperatures characteristic of these propellants require that an active cooling method be incorporated into the design. Studies¹ have indicated the necessity of using film cooling for missions requiring high delivered performance, repeatability of operation, insensitivity to duty cycle, and virtually unlimited periods of operation. The use of hydrazine fuels, such as monomethyl hydrazine (MMH), in conjunction with the high-energy earth- and space-storable oxidizers, has provided the basis for the incorporation of highly efficient cooling designs in rocket engines used in today's spacecraft.

The engine design concept results in low equilibrium wall temperatures, zero or minimum erosion in the combustor, and minimum soakback temperatures to other components during off periods. The key to this success is mainly due to the design of the film-cooling injector and the thermal design criteria used in the establishment of the over-all design.

This paper describes studies of eight propellant systems based on data obtained from the literature. Its accuracy was verified with test data for N_2O_4/MMH and F_2/MMH . Curves are presented which define the amount of film coolant required to maintain engine combustor temperature levels and system performance levels. These data methods can be used to provide design criteria for engines that are to be used in future space missions.

Typical Film-Cooled Rocket Engine for Precise Control

A typical film-cooled rocket engine for precise control is shown schematically in Fig. 1. The injector provides a high combustion efficiency gas core through propellant mixing and atomization. High combustion efficiency is attained in very short chamber lengths (L^* from 10 to 16 in. at low contraction ratios). The injector also provides a uniform, high-velocity fluid boundary layer at the combustion chamber wall with maximum penetration to the chamber throat.

The favored combustion chamber materials are molybdenum, columbium, and pyrolytic or composite graphite, which have high-temperature physical properties which result in thin-wall, light-weight designs.¹ Other designs² make use of more highly conductive chamber materials (Be or Cu) with thicker walls for increased heat transfer from the high heat flux throat region to the more highly film-cooled region, but they are heavier.

The propellant valves are close-coupled to the injector to minimize dribble volume (flow passage volume between the valve seat and the injector face). The dribble volume is minimized for precise and efficient minimum impulse control. In addition, the valves are thermally isolated to minimize the soakback temperature to the valve seat materials. The chamber seal provides a gas-tight interface between chamber and injector while providing high thermal resistance to limit heat soakback from the chamber to injector.

Film cooling, using N_2O_4 and hydrazine fuels, has been used in the reaction control rocket engines of the Lunar Orbiter, Apollo Service Module, and the Apollo Lunar Module. This technology has been extended to new chamber concepts applied to the higher-energy, fluorinated space-storable propellant combinations.

The radiation-cooled refractory chamber concept is presently used in most of today's long-life space engines. The use of fuel film cooling in these chambers, in conjunction with mixture ratio stratification by the main injector elements, has lowered the heat transfer to the combustor to such a degree that the wall temperatures encountered are well below the 2800° to $3000^\circ F$ structural and oxidation limit of the materials. For instance, in the advanced version of the Marquardt 100-lb-thrust R-4 engine, which has a molybdenum combustion chamber, the maximum wall temperature using N_2O_4/MMH is only $2600^\circ F$, whereas the engine is delivering a vacuum specific impulse (I_{sp}) in excess of 300 lbf-sec/lbm.

When fluorine-type oxidizers are used, molybdenum and columbium alloys cannot be used, because they are reactive with the combustion products. Pyrolytic graphite, in either free-standing or stacked-washer configurations, and composite carbon-fiber chamber designs, have a temperature capability far in excess of the refractory metals, with excellent strength up to 6000° to $6500^\circ F$. However, even with film cooling, which can lower the temperature of the combustion chamber walls, the oxidation of the walls can be a limiting factor in the cases where oxidizing combustion products are encountered. The use of film cooling also forms a chemically protective gas film which restricts the transport of the combustion products to the wall, thus reducing the rate of oxidation of the walls.

Analytical Methods

A portion of the fuel is injected along the chamber wall to act as an interface between the main core combustion gases and the combustor walls. In most of the rocket engine systems studied, there is a finite heat loss from the wall to the surroundings by radiation and conduction. In these studies, T_{aw} at any particular film-cooling rate is redefined and is called the "effective gas temperature" for nonadiabatic walls and is used in the equation to determine heat transfer due to convection between the combustion gases and the wall

$$Q/A = h_g(T_{eff} - T_w) \quad (1)$$

To determine film-cooling requirements, it was necessary to develop heat-transfer equations coupling the combustion gases to the coolant. The length of the liquid-cooled region is

$$L = \frac{\eta W_L C p_L (T_s - T_i)}{P h_g (T_r - T_s)} + \frac{\eta W_L \lambda}{P h_g (T_r - T_s)} \quad (2)$$

where the first term on the right represents the distance from the injection point for which the effective gas temperature varies from its initial injection temperature to its saturated liquid temperature, and the second term represents the dis-

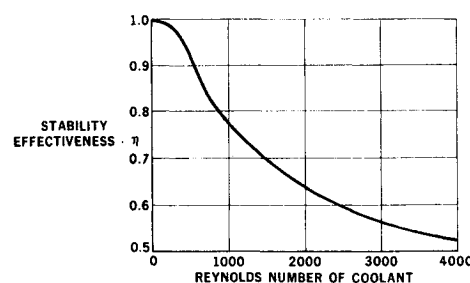


Fig. 2 Liquid film stability factor.⁴

tance over which the effective gas temperature is constant and equal to the coolant's saturation temperature.

Because of the liquid-film instability, the liquid coolant flow is not completely ideal.³ Therefore, the coolant flow was corrected by an efficiency factor that is a function of coolant Reynolds number. Recent work⁴ indicates a stability efficiency factor of the form shown in Fig. 2, where the Reynolds number of the coolant is $Re_L = W_L/\pi D\mu_L$.

The equations used to predict the effective gas temperature in the gaseous coolant region downstream of the liquid film region are

$$(T_r - T_{eff})/(T_r - T_w) = \exp(-h_g A / \phi C p_c W_c) \quad (3)$$

$$1/\phi = (Su_g/u_c)^{1/8} f(\beta) \quad (4)$$

where

$$\beta = u_g/u_c \quad (5)$$

and

$$f(\beta) = (1/\beta)^{-1.5(\beta^{-1}-1)} \quad \beta \leq 1.0$$

$$f(\beta) = 1 + 0.4 \tan(\beta - 1) \quad \beta > 1.0 \quad (6)$$

The slot height S is determined from the static density of the gas at the vaporization temperature and the velocity of the coolant at the original injection point u_c .

These equations are slight modifications of the equations developed by Hatch and Papell⁵ and were derived by using the following assumptions: 1) the coolant film exhibits no mixing or chemical reaction with the main core gas, 2) the coolant film temperature profile does not change rapidly as the coolant moves downstream, 3) the gradients across the coolant film are small, and 4) no heat transfer occurs in the chamber wall. Equations (4-6) were developed by Hatch and Papell to make their original equation agree with experimental data. The equations were programmed on the IBM 360 computer.

Heat-Transfer Coefficients (Convection)

Turbulent heat-transfer coefficients from the combustion gases to the film coolant and from the gaseous coolant to the wall were calculated using a modification of an equation presented by Bartz.⁴ Modifications were made in this equation, when required, to include the effect of the liquid and the gaseous films and the variations in transport properties between the main core gas and the coolant.

$$h_g = \frac{0.026\mu_f^{0.2}}{D^{*0.2}Pr_f^{2/3}} \left(\frac{W}{A^*}\right)^{0.8} \left(\frac{A^*}{A}\right)^{0.9} \left(\frac{H_r - H_w}{T_r - T_w}\right) \sigma \quad (7)$$

In small rocket engines, it has been shown that laminar flow of the combustion gases may occur due to favorable pressure gradients. Laminarization generally occurs if $Re_* < 200,000$ (based on throat diameter D_*). Laminarization may occur in the range $200,000 \leq Re_* < 400,000$, depending upon nozzle geometry and combustion effects. Laminar heat-transfer coefficients were determined using the method of Schoenman and Block.⁴ The static/total pressure ratio is determined as a function of distance upstream from the nozzle throat. From these data, the maximum logarithmic pressure gradient is determined and is extrapolated to the point where $P/P_0 = 0$, which determines the point upstream of the throat where the use of laminar heat-transfer coefficients begins. At this point, the laminar equation by Back⁸

$$St = 0.332Re_*^{1/2}Pr^{2/3} \quad (8)$$

is set equal to the turbulent Bartz equation. Thus, Re_* is defined, and the Stanton number can then be calculated further downstream. Schoenman and Block have concluded that this method provides good results for smooth-walled chambers with significant wall cooling and favorable pressure gradients. Test data at Marquardt on small rocket engines

have indicated that the method outlined previously provides good results.

The convective heat-transfer coefficient from the liquid film coolant to the chamber walls was derived assuming that there exists a linear velocity profile in the liquid film coolant. Then, the stress is $\tau_L = \mu_L u_L / \delta$, the flow in the coolant layer is $W_L = \pi r \rho_L u_L \delta$, and the friction coefficient is $C_f = 2\tau_g / \rho_g u_g^2$.

The modified Reynolds analogy⁹ for a turbulent boundary layer on a flat plate is given by

$$h_g Pr_g^{2/3} / \rho_g C p_g u_g = C_f / 2 = \tau_g / \rho_g u_g^2 \quad (9)$$

Combining the foregoing equations for τ_L and W_L and assuming that $\tau_g = \tau_L$ at the liquid-gas interface, then solving for u_L ,

$$u_L = (W_L u_g h_g Pr_g^{2/3} / \pi r \rho_L \mu_L C p_g)^{1/2} \quad (10)$$

The equation for heat transfer in a turbulent liquid flow on a flat plate¹⁰ is

$$h_L = 0.0288 Re_x^{0.8} Pr_L^{1/3} k_L / X \quad (11)$$

Combining Eqs. (10) and (11) and correcting the flow rate by the flow instability efficiency factor η gives

$$h_L = 0.0288 \frac{C p_L}{Pr_L^{0.667} \mu_L^{0.2} x^{0.2}} \left[\frac{\eta W_L u_g h_g Pr_g^{2/3} \rho_L}{\pi r C p_g} \right]^{0.4} \quad (12)$$

for the heat-transfer coefficient between the liquid film and the engine walls. This equation has been found to give best results if the temperature difference in calculating the heat flux is taken between the average film temperature T_f and the engine wall temperature T_w .

Performance Losses Due to Film Cooling

When a boundary-layer film of liquid or gas flows through a rocket exhaust nozzle throat at a temperature different from the main bulk of the exhaust gases, the net thrust of the engine will be less than that which would result if the gases had been thoroughly mixed. This gas stratification is essentially independent of the chemical combustion efficiency. Studies have been conducted to determine the performance when film cooling is used in rocket engines. Two models of the combustion process have been evaluated: complete mixing of the flow streams with thermal equilibrium but no reaction; and no mixing of the flow streams, but the thrust of the coolant is considered. These two methods of predicting performance were examined, and the second model was chosen for this study, based on the ease of applicability and the close correlation of this method with the test data. The equation used in this study was:

$$I_{sp} = (WI_{sp})_g + (WI_{sp})_c / (W_g + W_c)$$

This mass-flow averaging is appropriate if the γ 's are equal for the flow streams. This condition is equivalent to the assumption that the two flows have equal Mach numbers at each flow station. Although this condition is not strictly met, no significant inaccuracies are introduced by assuming that γ is constant for all gas portions.¹¹ The performance of the main core gas and the coolant is based on thermochemical equilibrium with injection conditions as the initial condition.

Experimental Results

Two engine design concepts were tested using N_2O_4 /MMH at the 100-lb thrust level and F_2 /MMH at the 1000-lb thrust level. For the former R-4 series, experimental data from two series of tests are shown in Fig. 3.

The combustion chambers were made of molybdenum. The injector used consists of a basic set of eight unlike doublets. The R-4D engine incorporates a preigniter for ignition overpressure suppression, and eight film-cooling holes for

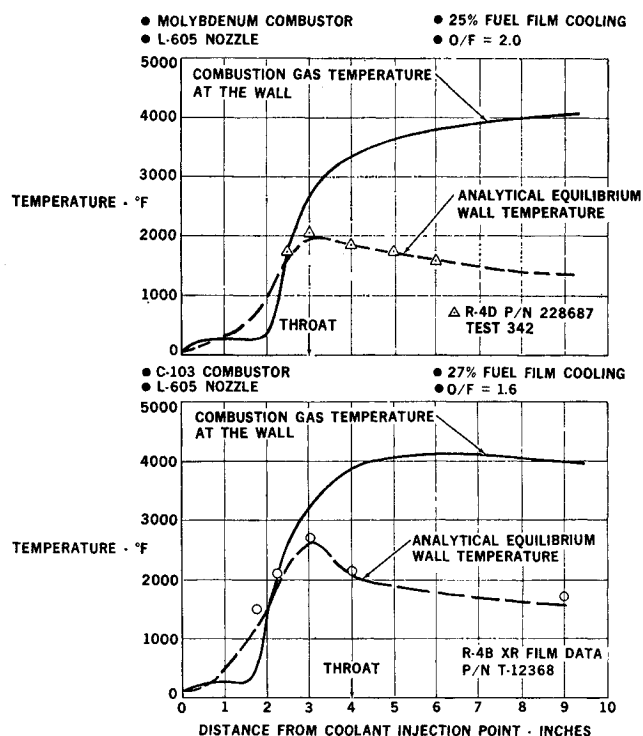


Fig. 3 Temperature characteristics for R-4D (top) and R-4B (bottom) engines: propellant combination N_2/O_4 -MMH; $P_c = 97$ psia; vacuum, $A/AT = 40$; thrust = 100 lb.

chamber cooling. The R-4B engine consists of eight unlike main doublets and 8 or 16 cooling holes for chamber cooling. These injectors are shown in Fig. 4. In both cases, approximately 25% of the total available fuel was used to cool the combustor.

With the refinement of the film-cooling injection techniques, significant reductions in chamber temperatures were accomplished while increasing the performance to equal that obtained without film cooling. The primary injection core consists of multiple unlike doublets, which have been optimized in terms of Rupe number and momentum angle to maximize the performance while maintaining the low mixture ratio stratification zone in contact with the film coolant. The fuel film coolant flows from circumferential fuel-cooling holes, which are directed at various chamber impingement angles in order to insure that a complete coverage of the walls is maintained and the cooling efficiency is maximized. The performance obtained from this series of engines is shown in Fig. 5. An increase in performance was obtained by optimizing the main core while maintaining essentially an identical film-cooling rate. The method for predicting the performance of this engine, based on the methods described in the previous section on Analytical Methods, has proven adequate.

The applicability of film cooling using the F_2/MMH propellant combination has been demonstrated with complete success using a thin-walled, radiation-cooled, pyrolytic graphite chamber. Figure 6 compares the predicted and

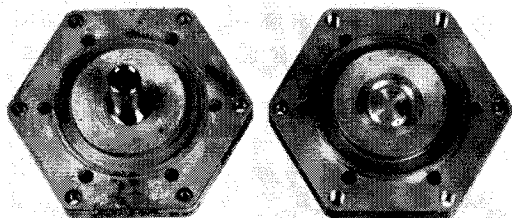


Fig. 4 100-lb thrust earth-storable propellant injectors.

measured temperature distributions for this chamber using the injector shown in Fig. 7. The injector consists of 44 like-on-like doublets, with 16 film-cooling holes. The predicted performance of this injector concept is shown in Fig. 8, in terms of I_{sp} . The performance obtained is well within the accuracy of the values predicted for the propellant combination.

Film-Cooling Requirements and Study Results

The film-cooling model developed and correlated with engine test data was used to determine film-cooling requirements for, and compare the film-cooling effectiveness of, eight propellant combinations for the parameters shown in Table 1. In all cases, the fuel was used as the film coolant since the fuel was found to have a significant advantage in the area of cooling properties and would, in most cases, be nonreactive with the combustor structure. The results of the study are shown in Fig. 9 for the 100-lb-thrust, 100-psia chamber pressure level. Each plot shows the required amount of coolant needed to maintain various effective gas temperatures at the throat as a function of mixture ratio.

In all of the studies conducted, it was assumed that the combustion efficiency was 100% and that the chemical composition was in shifting equilibrium to the throat and was frozen thereafter. The effective gas temperature at the throat T_{*eff} was chosen as the characteristic temperature, since maximum heat transfer to the combustor walls usually occurs at or near the throat. For engine systems that are insulated, the wall temperature will be nearly identical to T_{*eff} , whereas in radiation- or conduction-cooled engines, rapid evaluation of the wall temperature can be made using conventional heat-transfer methods. These data were generated using a Fortran IV computer program, which rapidly calculates T_{*eff} , heat-transfer coefficients, and flow characteristics as a function of distance from the injection point. The L^* (combustor volume/ A_*) and contraction ratio were minimized to reduce the total heat load to the system while maintaining the highest performance level from a combustion efficiency standpoint.

As a rule, the higher performance propellant systems have higher combustion gas temperatures. Probably the most important aspect of film cooling is the engine mixture ratio (oxidizer/fuel, O/F). A decrease in O/F (below the optimum) results in a decrease in the flame temperature and an increase in the relative amount of film coolant available for cooling. Since the fuel density usually is lower than the oxidizer density, this decrease also results in an increase in system weight due to the loss in propellant bulk density.

The most desirable condition for film cooling is to minimize the total heat transfer to the coolant by essentially minimizing L^* . In most cases, however, there is a limit to this decrease, since the combustion efficiency is a function of L^* . In addition, the chamber-pressure level will significantly affect the film-cooling requirements since the surface area, i.e., heat-

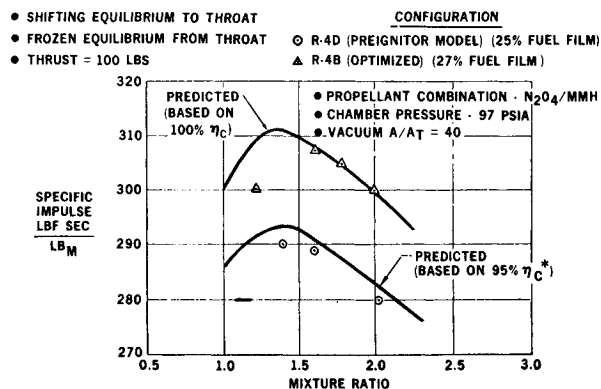


Fig. 5 100-lb thrust engine performance.

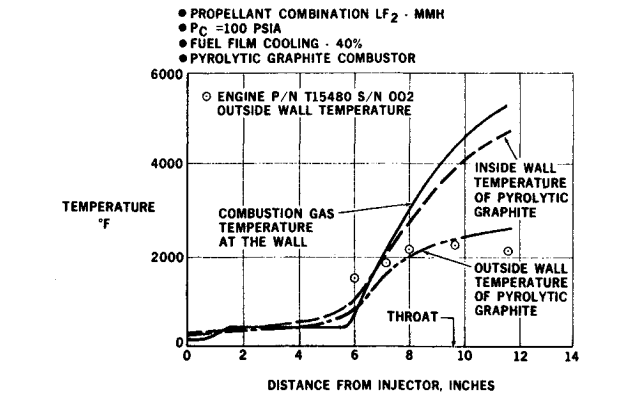


Fig. 6 1000-lb thrust engine steady-state temperature characteristics.

transfer load and the chamber-pressure level, is not linear for a constant L^* .

Some of the results of the studies conducted to determine the film-cooling requirements of eight propellant combinations are presented in Fig. 9. The N_2O_4/MMH system exhibits excellent adaptability to film cooling at all thrust levels and chamber pressures investigated. The wall temperatures were easily controlled to values low enough to allow the use of conventional chamber materials (molybdenum, tantalum-tungsten, stainless steel, columbium, etc.). Only slight performance degradation was experienced from the film cooling allowing the engine to operate at the optimum or near-optimum mixture ratio. The ClF_3/MMH system, even though operating at higher gas temperatures than N_2O_4/MMH , also permits film cooling to temperatures low enough to use conventional chamber materials, while operating at nearly optimum O/F.

The F_2/H_2 combination displays high wall temperatures at almost all conditions investigated ($>4000^\circ F$). It also exhibits appreciable performance losses with increasing amount of film cooling and/or variations from the optimum O/F for performance. These conditions strongly suggest the necessity for using advanced chamber materials for this propellant system if the high performance inherent to this combination is to be used advantageously. Special consideration also should be given to the necessity of running F_2/H_2 at high enough mixture ratios to minimize the system weight degradation which would result from carrying large amounts of hydrogen, further suggesting the need for high-temperature chamber materials.

The O_2/H_2 system shows excellent adaptability to film cooling. Through the use of fuel-rich mixture ratios and the appropriate amount of film cooling, wall temperatures can be kept low enough to allow the use of conventional chamber materials. However, in order to operate at this low wall temperature, the propellant bulk density and deliverable I_{sp} decrease, resulting in greater system weight.

Table 1 Design conditions for film cooling studies			
Propellant combinations	Chamber pressure, psia	Thrust level, lb	Chamber dimensions
N_2O_4/MMH	10	10	$L^* = 10$ in. $A_c/A_t = 4$
ClF_3/MMH		100	
F_2/MMH		1000	
OF_2/MMH	100	10	$L^* = 15$ $A_c/A_t = 2$
F_2/H_2		100	
O_2/H_2		1000	
OF_2/B_2H_6	500	1000	$A_c/A_t = 2$
OF_2/CH_4		1000	

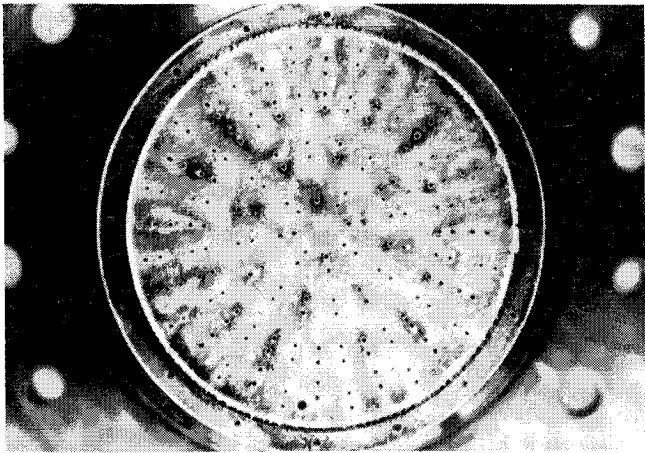


Fig. 7 1000-lb thrust space-storable propellant injector.

The OF_2/MMH system does not permit film cooling to temperatures less than $3000^\circ\text{--}4000^\circ F$, except at high thrust/chamber-pressure ratios ($F/P_c > 10$); however, the sensitivity of deliverable I_{sp} to the percentage of fuel used for film cooling is not as great as that experienced with the F_2/H_2 and O_2/H_2 combinations, because the decomposition of the MMH contributes to the system performance. The F_2/MMH combination is almost identical in characteristics to the OF_2/MMH combination, except that wall temperature are slightly higher for similar operating conditions.

The OF_2/B_2H_6 and OF_2/CH_4 combinations are not suited to film cooling over the ranges of F and P_c examined. Film cooling flows up to 50% of total flow, and fuel-rich operation do not seem to alleviate the severe wall temperatures ($>4000^\circ F$ in almost all cases).

Concluding Remarks

Propellant combinations using one of the hydrazine-type fuels are well suited to film cooling. These fuels have large heats of vaporization, so that they remain in the more efficient liquid state for a longer time. In addition, the optimum mixture ratios for systems using these fuels is low compared with systems using hydrogen, LPG's, and diborane. This advantage is offset to some degree by the lower I_{sp} of a systems using one of the hydrazine fuels. Hydrogen and methane (and similar LPG's) do not have very large heats of vaporization. They do have high specific heats in the gaseous phase, which result in fairly good cooling, but in most cases, this favorable characteristic is negated by the high mixture ratios at which systems using these fuels must operate. Diborane has fairly large values of heat of vaporization and gaseous specific heat, but high mixture ratios and the high combustion temperatures associated with use of this fuel, and its reactivity with almost all materials is a drawback.

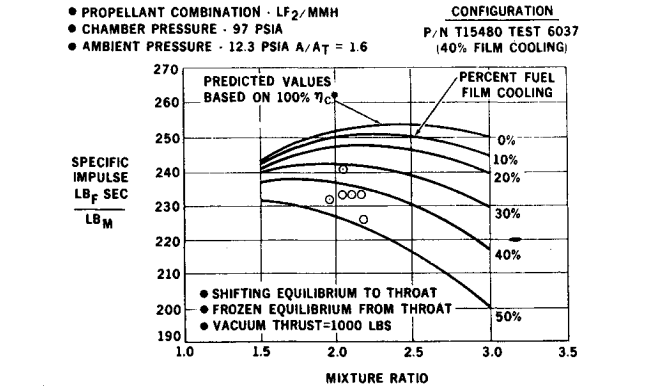


Fig. 8 1000-lb thrust engine performance.

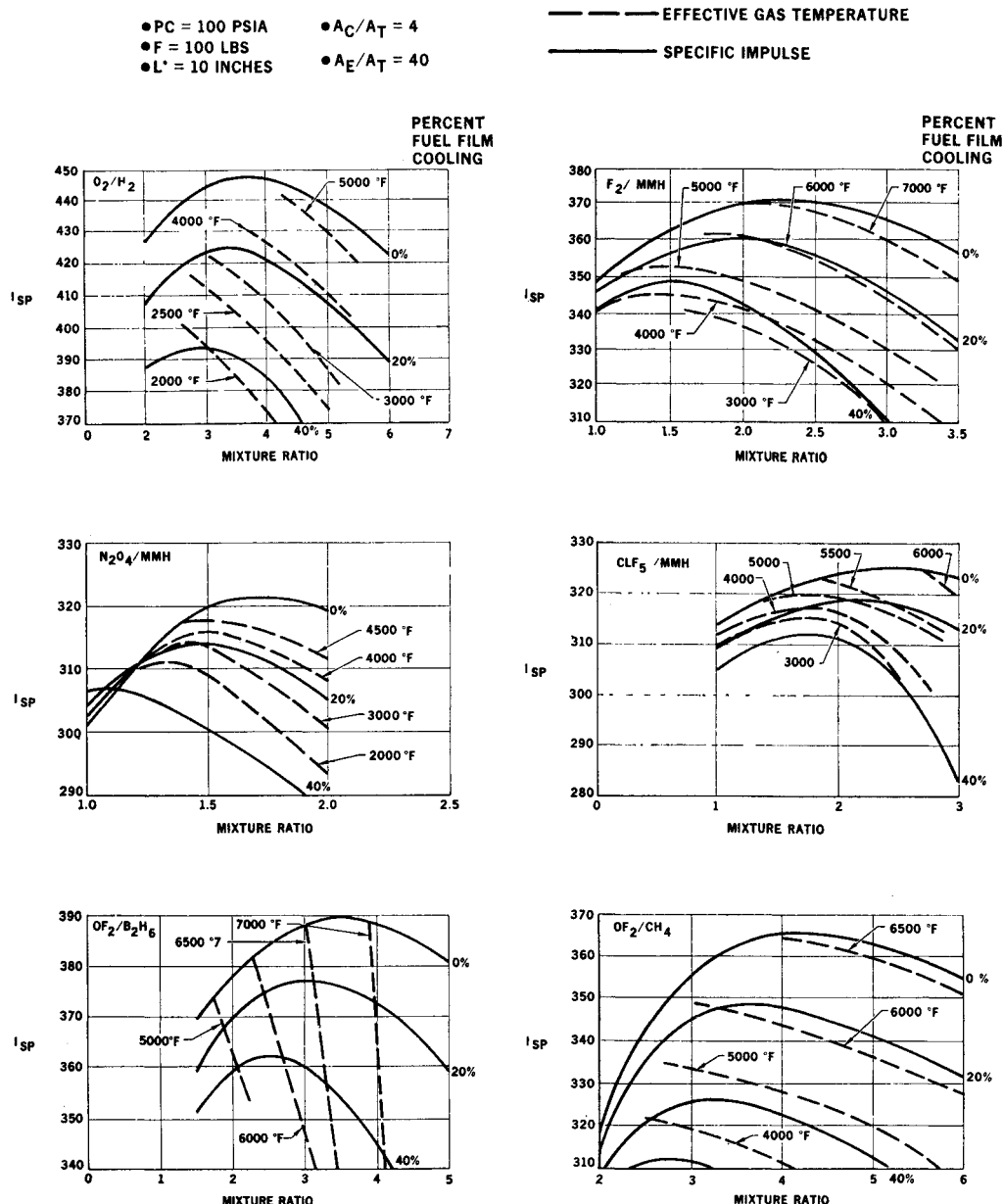


Fig. 9 Film-cooling requirements.

In general, film cooling is most efficient when the F/P_c ratio is maximized. In addition, a lower P_c means a lower cooling requirement; however, this thermal advantage is offset by a loss in I_{sp} . Therefore, each engine/propellant system combination must be individually analyzed to determine the optimum operating conditions.

In general, then, earth-storable propellant combinations such as N₂O₄/MMH and CLF₅/MMH, are readily adaptable to use in small film-cooled spacecraft engines. The space-storable propellant systems such as OF₂/MMH, OF₂/B₂H₆, OF₂/CH₄, or F₂/MMH, and deep-cryogenic systems such as O₂/H₂ and F₂/H₂, will require both film cooling and advanced, high-temperature chamber materials if their potentially high I_{sp} s are to be realized at satisfactory system weight.

References

- Campbell, J. G. et al., "Thrust Chamber Cooling Techniques for Spacecraft Engines," CR-50959, NASA.
- Campagna, F. E., "Development and Demonstration of a Beryllium Rocket Engine," AFRPL-TR-66.25, Vol. 1, Contract AF05(611)-11393, Nov. 1966, Rocketdyne, Div. of North American Rockwell, Inc., Canoga Park, Calif.
- Graham, A. R., "An Experimental and Theoretical In-

vestigation of Film Cooling of Rocket Motors," Rept. F 57-3, Oct. 1957, Rocket Lab., Purdue Univ., Lafayette, Ind.

⁴ Glenn, L. A. and McFarland, B. L., "Advanced Experimental Thrust Chamber Program (Phases I and II), Final Report," Rept. RO6857, Dec. 1966, Rocketdyne, Canoga Park, Calif.

⁵ Hatch, J. E. and Papell, S. R., "Use of a Theoretical Flow Model to Correlate Data for Film Cooling or Heating an Adiabatic Wall by Tangential Injection of Gases of Different Fluid Properties," TN D-130, Nov. 1959, NASA.

⁷ Schoenman, L. and Block, P., "Application of Laminar Boundary Layer Heat Transfer Theory to Low Thrust Rocket Nozzles," Paper 67-447, July 1967, AIAA; also *Journal of Spacecraft and Rockets*, Vol. 5, No. 9, Sept. 1968, pp. 1082-1089.

⁸ Back, L. H. and Witte, A. B., "Prediction of Heat Transfer from Laminar Boundary Layers, with Emphasis on Large Free-stream Velocity Gradients and Highly Cooled Walls," *Journal of Heat Transfer*, Vol. 88, Ser. C, No. 3, Aug. 1966, pp. 249-256.

⁹ Hill, P. G. and Peterson, C. R., *Mechanics and Thermodynamics of Propulsion*, Addison-Wesley, Reading, Mass., 1965, pp. 1-97.

¹⁰ Holman, J. P., *Heat Transfer*, McGraw-Hill, New York, 1963, pp. 1-116.

¹¹ Wrobel, J. R., "Some Effects of Gas Stratification in Choked Nozzle Flows," *Journal of Spacecraft and Rockets*, Vol. 2, No. 6, Nov.-Dec. 1965, pp. 918-922.